

Write a Python Program to Implement Feed Forward Neural Network

Aim

To develop a Python program that implements a Feed Forward Neural Network (FFNN) for classification.

Objective

- To understand the architecture of Feed Forward Neural Networks.
- To create and train a neural network.
- To perform classification using TensorFlow/Keras.
- To obtain predictions for the given input data.

Theory

A **Feed Forward Neural Network (FFNN)** is the simplest form of Artificial Neural Network in which data moves only in the forward direction—from the input layer to the hidden layer and finally to the output layer. Unlike recurrent neural networks, FFNNs do not have feedback connections. During training, the network learns by adjusting its weights through optimization algorithms. FFNNs are commonly used in pattern recognition, classification, and prediction problems.

Algorithm

Step 1: Import TensorFlow and NumPy libraries.

Step 2: Create the input and output datasets.

Step 3: Build the neural network model.

Step 4: Add input, hidden, and output layers.

Step 5: Compile the model.

Step 6: Train the neural network.

Step 7: Predict the output for the input data.

Step 8: Display the predicted values.

Flowchart

Start

|



Import Libraries



Create Dataset



Build Neural Network



Compile Model



Train Network



Predict Output



Display Results



Stop

Python Program

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense
```

```
import numpy as np
```

```
X = np.array([[0,0],
```

```
[0,1],  
[1,0],  
[1,1]])
```

```
Y = np.array([[0],  
[1],  
[1],  
[0]])
```

```
model = Sequential()
```

```
model.add(Dense(4, input_dim=2, activation='relu'))
```

```
model.add(Dense(1, activation='sigmoid'))
```

```
model.compile(optimizer='adam',  
              loss='binary_crossentropy',  
              metrics=['accuracy'])
```

```
model.fit(X, Y, epochs=200, verbose=0)
```

```
prediction = model.predict(X)
```

```
print(prediction)
```

Sample Output

```
[[0.02]  
[0.95]  
[0.94]  
[0.03]]
```

Result

The Python program successfully trained the Feed Forward Neural Network and generated predictions for the given dataset.

Conclusion

The Feed Forward Neural Network successfully learned the input patterns and produced accurate predictions. This experiment demonstrates the basic working principle of Artificial Neural Networks and their application in classification problems.

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

X = np.array([[0,0],
              [0,1],
              [1,0],
              [1,1]])

Y = np.array([[0],
              [1],
              [1],
              [0]])

model = Sequential()

model.add(Dense(4, input_shape=(2,), activation='relu'))
model.add(Dense(1, activation='sigmoid'))

model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

model.fit(X, Y, epochs=200, verbose=0)

prediction = model.predict(X, verbose=0)

print(prediction)
```

```
... [[0.53196716]
      [0.59143835]
      [0.4883123 ]
      [0.3477211 ]]
```